Homework 06

Robotic Motion

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1. Question 13

Question: Finding periodic motions.

1.1. Q13-A

Verify, using numerical solutions, that for $\ddot{\vec{r}} = -\frac{k}{m}\vec{r}$ that for any and all initial conditions you get periodic motions (straight lines, circles, or elipses). Extra credit, verify this with pencil and paper. This is a 'linear' central force'.

The given ODE is similar to the equation $\ddot{x} = -\lambda x$ which yields the general solution of $x(t) = A\sin(\lambda t) + B\cos(\lambda t)$ for any constants A, B. $\therefore$ the solution to $\ddot{\vec{r}} = -\frac{k}{m}\vec{r}$ must also be periodic in nature for any given k, m.

To verify this, I added the constraint in the `fsolve` objective function that the terminal and initial states must all be the same.

```
function error = periodicResidualAllStates(all_init, problemODE, p)
    z0 = all_init(1:4);
    T = all_init(5);

    RHS = @(t, z) problemODE(z, p);
    sol = ode45(RHS, [0 T], z0);

    z_T = deval(sol, T);
    error = [
        z0 - z_T; % Return back to same point
        p.z0(1) - z_T(1); % Enforce x0_converged = x0
        p.z0(2) - z_T(2); % Enforce y0_converged = y0
        p.z0(3) - z_T(3); % Enforce vx0_converged = vx0
        p.z0(4) - z_T(4); % Enforce vy0_converged = vy0
        % Use p.z0 as all_init passed into function changes as algo.
        % converges
    ];
end
```

Listing 1: fsolve Objective Function

For random initial conditions the periodic solution were found, and the norm of the error was computed and shown below in the results.

z0_1	z0_2	z0_3	z0_4	zp_1	zp_2	zp_3	zp_4	T	ErrorNorm
2.8847	-0.53752	0.3409	3.9768	2.885	-0.53757	0.34093	3.9771	6.283	0.00044594
7.2342	0.010207	2.6384	-0.19151	7.2348	0.010208	2.6386	-0.19152	6.283	0.00068327
3.148	3.2	-2.0843	2.3742	3.1483	3.2003	-2.0845	2.3744	6.283	0.00050193
5.0055	1.4615	9.6492	-2.0428	5.006	1.4617	9.6501	-2.043	6.283	0.0010327
-1.6853	-7.7494	5.5524	5.3354	-1.6854	-7.7501	5.5529	5.3359	6.283	0.00098077
1.6005	5.4924	1.9185	2.454	1.6007	5.4929	1.9187	2.4542	6.283	0.00058762
2.5647	3.1998	1.5083	14.937	2.565	3.2	1.5084	14.938	6.283	0.0013862
-3.8333	-8.2715	-4.7034	-4.4667	-3.8337	-8.2723	-4.7038	-4.4671	6.283	0.001026
-0.16805	0.15765	-0.092668	3.7067	-0.16806	0.15766	-0.092676	3.707	6.283	0.00033068
3.9463	4.7561	9.8913	7.1153	3.9467	4.7566	9.8922	7.1159	6.283	0.0012213

Figure 1: Results

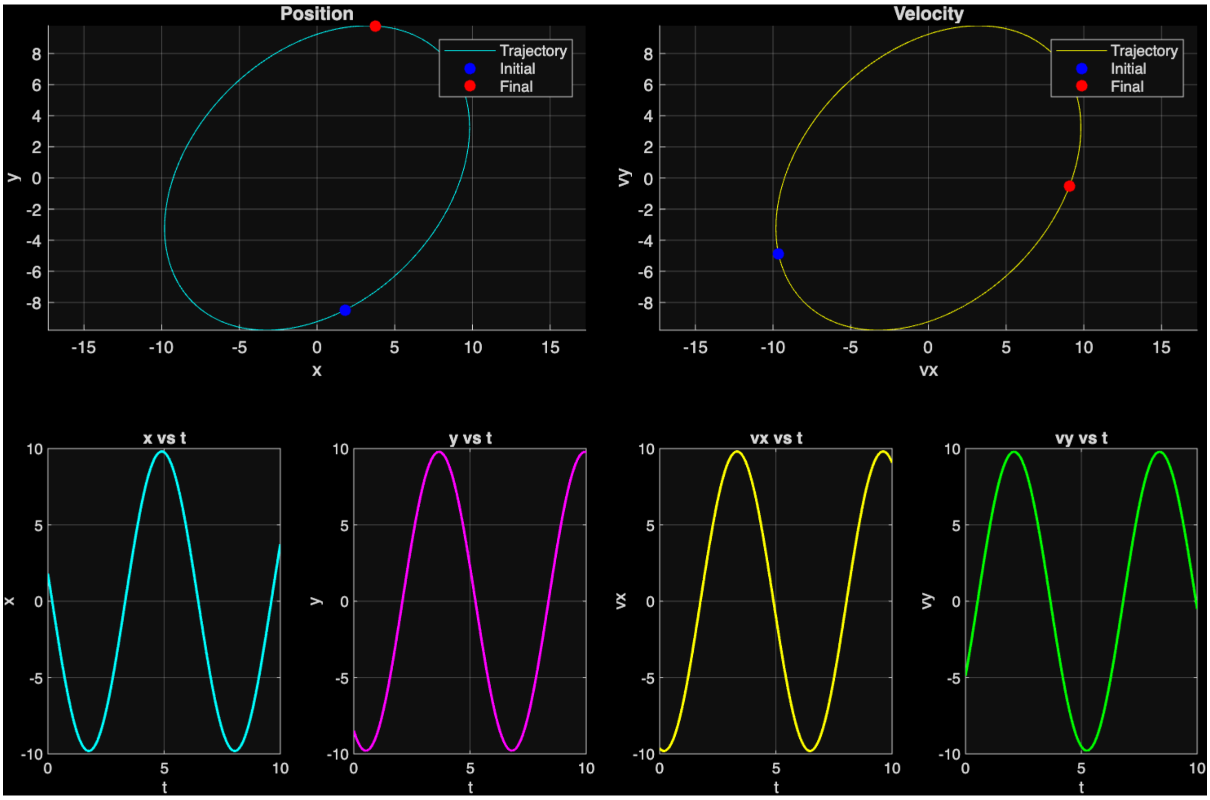

Figure 2: A Solution

1.2. Q13-B

System:

$$\ddot{\vec{r}} = -\frac{GM}{r^3}\vec{r}$$
$$= -\frac{GM}{r^3} r\hat{r}$$
$$= -\frac{GM}{r^2}\hat{r}$$
$$\Rightarrow F \propto \frac{1}{r^2} \therefore \text{inverse square law}$$

Choosing $G = M = 1$ and sampling points around the state $\vec{r}_0 = [1, 0]\vec{v} = [0, 1]$, the same approach as the previous question is used to find out if solutions are periodic or not.

z0_1	z0_2	z0_3	z0_4	zp_1	zp_2	zp_3	zp_4	T	ErrorNorm
1.2862	−0.68617	−0.81331	1.7934	1.7402	0.42796	−0.14955	0.68704	24.762	1.7641
0.61805	−1.3712	0.010307	1.2041	0.61805	−39.699	−2.5511e−05	0.0008952	5.8736e−08	38.347
0.58897	0.66356	0.22578	0.78716	0.589	0.66356	0.22578	0.78714	25	2.8266e−05
0.95597	0.45818	−0.44142	−0.054947	0.95597	0.45818	−0.44142	−0.054947	25	0
0.84442	0.12915	0.50943	0.96988	0.84817	0.085712	0.5391	0.95493	25	0.054815
0.54264	0.59634	−0.11353	1.807	0.54512	2.3549	−0.42891	−0.090122	24.537	2.606
0.9102	−0.0062921	−0.091861	0.078762	0.9102	−0.0062921	−0.091861	0.078762	25	0
0.073048	−0.96122	1.7848	0.79981	0.0856	−2.9793	−0.13314	−0.14445	24.827	2.9399
1.9404	0.34916	1.8593	1.9271	2.0004	1.2637	0.28184	0.20951	24.525	2.5057
−0.22696	−0.32724	0.89165	1.2882	NaN	NaN	NaN	NaN	NaN	NaN
3.2652	−0.04789	−1.5519	1.4441	3.2652	29.116	−0.0042134	−0.011304	−6.1793e−12	29.241
0.088181	0.049435	1.078	1.3082	NaN	NaN	NaN	NaN	NaN	NaN
1.2996	−0.19722	−0.14642	0.89693	1.3016	−0.3487	−0.35133	0.85561	24.971	0.25816
−1.799	0.39328	0.99025	−0.29763	−1.804	2.2376	0.15233	−0.53065	25.318	2.0391
−0.52197	0.62101	−1.5075	−0.6794	−0.51971	2.6414	0.043495	0.32913	24.471	2.7395
1.789	−0.65428	1.2449	−0.29226	1.7862	−1.5714	0.43437	−0.53778	28.763	1.2483
0.38565	0.2417	0.54935	1.4676	0.41803	0.31593	0.28924	1.3831	24.993	0.28524
1.1915	−0.2298	−0.57922	1.4805	1.3858	0.52749	−0.17259	0.83206	24.761	1.0941
0.61317	0.42162	1.0877	−1.2493	NaN	NaN	NaN	NaN	NaN	NaN
2.8045	−0.63206	1.3165	2.5516	2.8047	−0.45789	0.22361	0.12713	24.554	2.6651

Figure 3: Results

All entries with T close to 0 or NaN are invalid. One of these invalid and valid solutions is shown below.

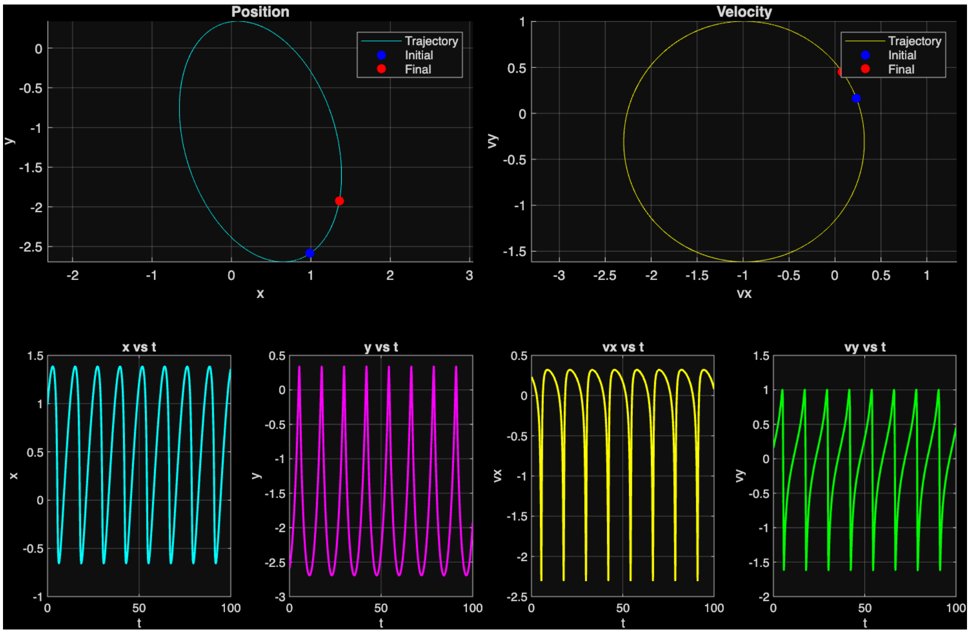

Figure 4: Q13-B: A Periodic Solution

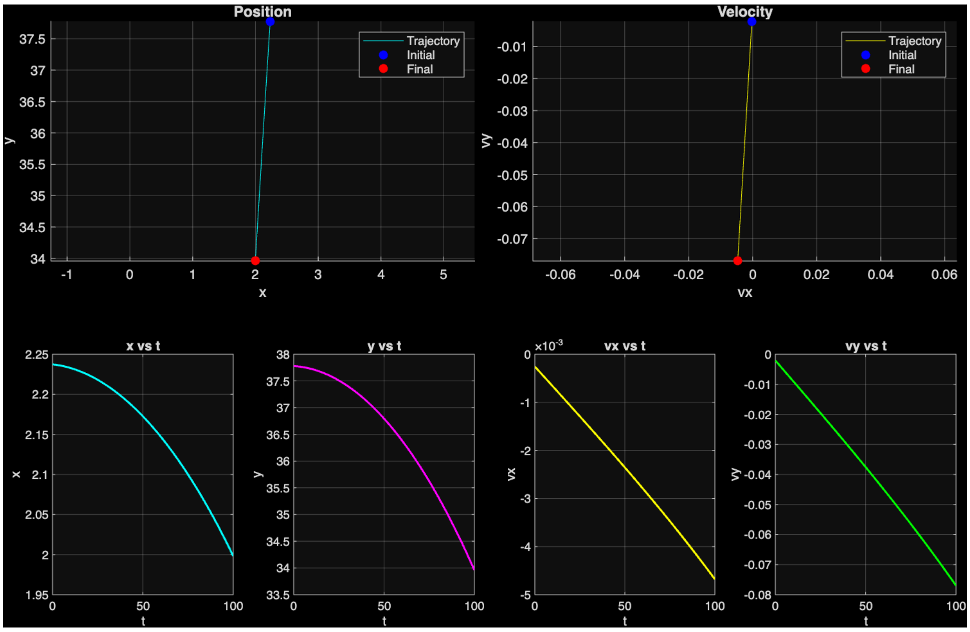

Figure 5: Q13-B: A Non Periodic Solution

2. Q13-C

System:

$$\ddot{\vec{r}} = -kr^{n-1}\vec{r}$$
$$k > 0 \quad \& \quad n \neq 1, -2$$

I selected $k = 1$ and $n = 2$

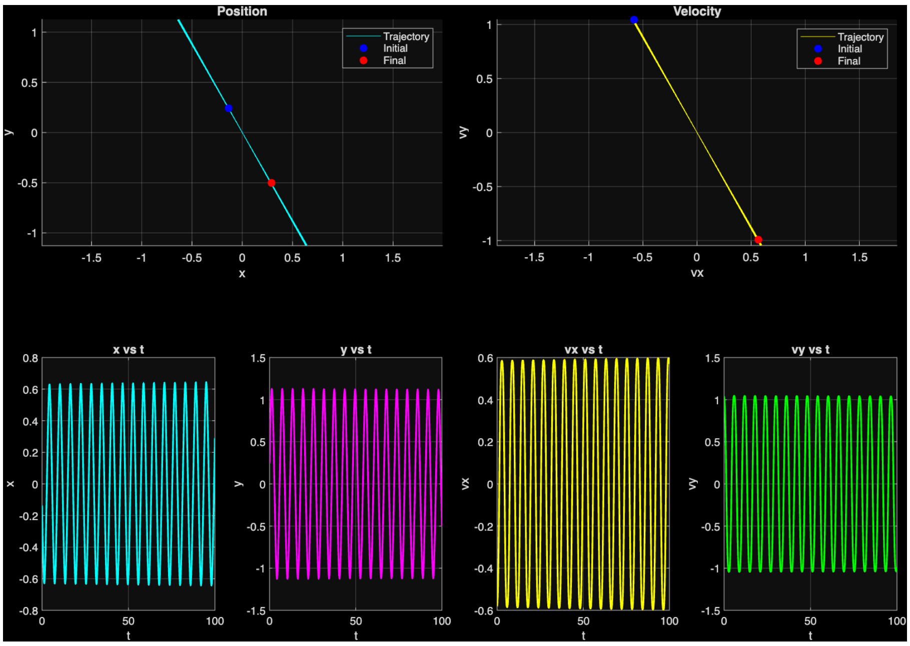

Figure 6: Straight Line Motion towards origin if we start at rest

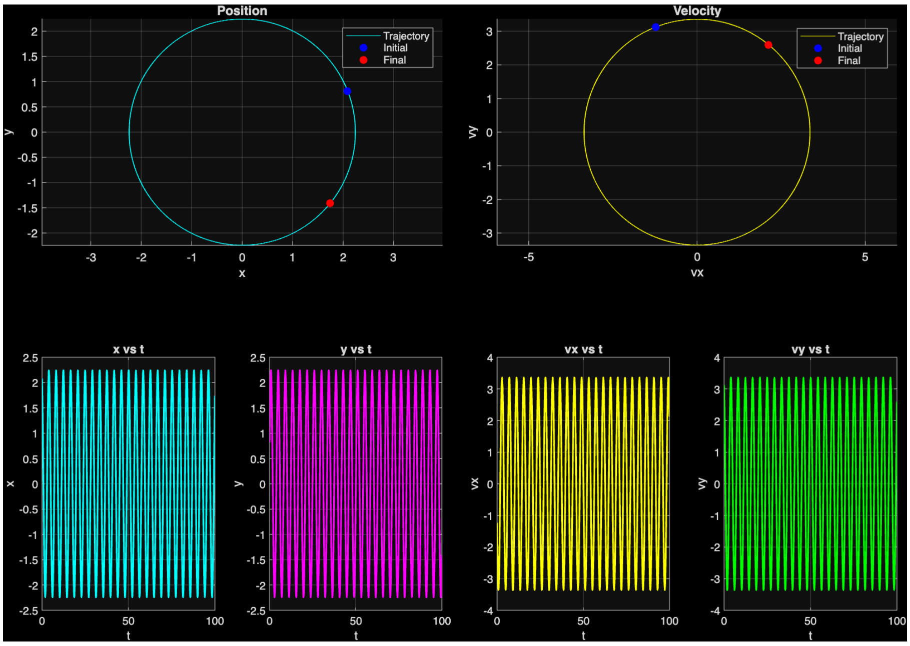

Figure 7: Circular motion if v_y != 0

3. Q13-D

Show that for the above problem, for random initial conditions you do not generally get periodic motions.

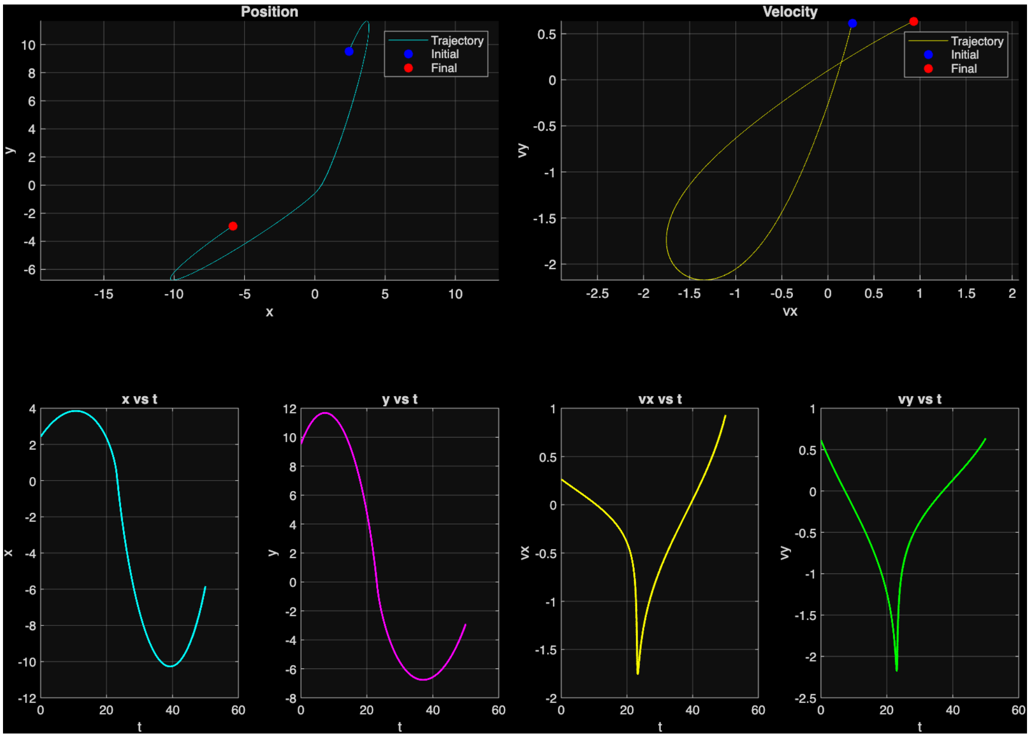

Figure 8: n=-1, random z0

We can easily see this from the error table I have been making.

z0_1	z0_2	z0_3	z0_4	zp_1	zp_2	zp_3	zp_4	T	ErrorNorm
2.7694	1.3832	0.017864	−1.0045	2.7697	2.5939	0.68424	−0.73052	23.837	1.4088
2.5331	1.5083	1.4789	1.2628	2.5331	7.4744	0.097114	0.75865	−3.0561e−14	6.1448
0.91787	1.2838	0.31999	−0.061367	0.84896	1.3235	0.3428	−0.055352	25.132	0.082952
0.5441	1.8002	1.0751	1.5036	0.5441	29.366	−0.0059513	−0.086299	2.3959e−10	27.632
3.611	−0.43724	1.543	0.48724	3.611	2.5122	0.39671	0.30759	1.4292e−16	3.1695
3.0143	1.8077	1.7205	−0.31761	3.0148	−0.016996	−0.005715	−1.0008	−18.938	2.6031
1.9735	0.96414	4.0585	1.7266	1.9735	3.2399	−0.67338	0.40696	−7.1409e−16	5.414
4.4206	1.1145	1.1118	1.1132	4.4206	8.0475	0.095693	0.65111	1.1086e−14	7.0223
1.5722	1.9416	0.74237	−0.8736	1.5724	3.5384	0.91462	−0.40641	24.323	1.6726
2.9708	−0.86853	−0.94425	0.78841	2.9712	−2.4111	0.63076	0.77718	24.036	2.2046
1.1491	−0.21073	0.73805	0.011098	1.1468	−0.20871	0.73823	0.011016	25.578	0.0030657
1.8772	0.97207	−0.26853	−0.72423	1.8776	−2.2636	−0.77035	−0.63903	18.473	3.2754
0.0005213	0.30976	1.5643	−0.20226	0.00052432	0.30976	1.5643	−0.20226	25	3.1641e−06
2.6013	1.2403	1.8908	1.4894	2.6013	7.8189	0.24661	0.60446	−1.1759e−14	6.8384
1.4112	−0.15684	0.94839	−0.16485	1.4111	−0.15685	0.94841	−0.16485	25	6.2322e−05
3.5527	2.0798	0.99514	0.072873	3.5531	2.0415	0.49869	−0.8678	25.741	1.0643
2.6108	0.043726	0.77498	−0.030467	2.5731	0.043833	0.77954	−0.030483	25	0.037973
2.2056	1.8284	1.6425	1.6436	2.2056	14.292	−0.098962	0.35523	1.7854e−14	12.651
1.1853	0.8798	−0.083348	0.88483	1.192	0.63225	−0.46436	0.89187	25.358	0.45448
2.4257	1.9963	1.9992	1.1532	2.4257	9.5251	0.26387	0.6153	−2.351e−13	7.7449

The error term is sometimes close to zero (periodic motion) and sometimes way off (non periodic). The above figure is from the last entry of the table with an error of ≈ 7.75.

4. Q13-E

$$\ddot{\vec{r}} = -kr^{n-1}\vec{r}$$
$$k > 0 \quad \& \quad n \neq 1, -2$$

Find a non circular and non straight line periodic motion:

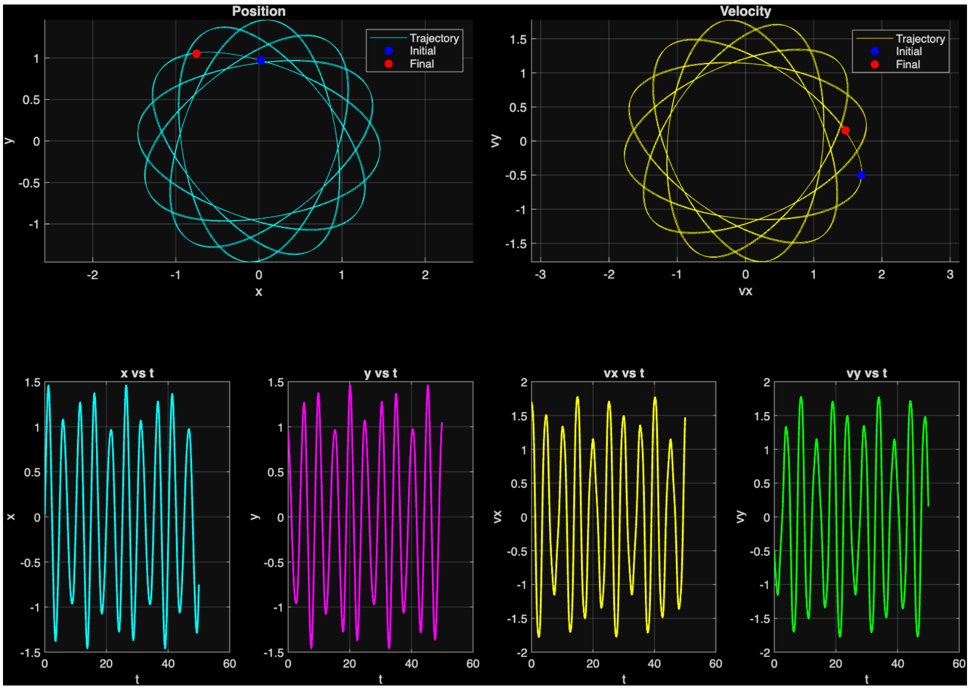

Figure 9: n=3, k=1, r0 = [1; 0], v0 = [2; −0.5]

What happens if $n \leq -3$?
- I didn't notice any periodic motions from what I tested with.
- I think this is because at $n \leq -3$, the force drops off very quickly and is not enough to pull the mass back in.

- Asking Gemini, it gave me a slightly complicated explaination involving the potential field of the force and the angular momentum of the system which I didn't fully understand.

5. Q13-F

Find periodic motions in the two spring mass damper system with g=0, all damping=0, and anchors not vertically aligned

The error term for all configurations I tried was too large to give periodic motion even if I increased the number of points. I am not sure if that is due to a bug in my rootfinding.